

FINAL PROJECT REPORT

PHISHING AND SCAM DETECTION

Internship Final Week – Week 06

Submitted By

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Domain

Digital Marketing

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Machine Learning Based Detection of Suspicious Digital Messages

1. Abstract

Phishing and online scams are common forms of digital fraud in which users are encouraged to provide sensitive information, click suspicious links, or respond to misleading messages. This project focuses on developing a basic machine learning-based Phishing and Scam Detection System that can classify digital messages as legitimate or suspicious. The system uses characteristics such as suspicious keywords, urgency-related language, links, message length, promotional wording, unusual URL patterns, excessive punctuation, and abnormal capitalization. Different classification algorithms, including Logistic Regression, Decision Tree, and Naive Bayes, were studied and compared. The final stage of the project focused on selecting a suitable model, testing the complete workflow, integrating the prediction process into a simple application interface, and completing the project documentation. The project demonstrates how machine learning can support safer digital communication and improve awareness of suspicious content.

2. Introduction

Digital communication and online marketing have become an important part of everyday activities. Businesses use email, messages, websites, social media, and other digital channels to communicate with customers. At the same time, scammers can misuse these channels to distribute phishing links, fake offers, fraudulent promotions, and misleading messages. Because suspicious content can appear similar to genuine marketing communication, identifying it manually can be difficult.

The Phishing and Scam Detection project applies machine learning techniques to this problem. Instead of depending on one keyword, the system considers multiple characteristics of the input and uses a trained classification model to generate a prediction. The project was developed as a practical learning exercise combining digital marketing concepts with machine learning and cybersecurity awareness.

3. Problem Statement

Users receive a large number of digital messages and promotional communications. Some may contain suspicious links, urgent requests, misleading offers, or other scam-related patterns. Manual identification is time-consuming and users may not always recognize these warning signs. Therefore, there is a need for a simple automated system that can analyze message characteristics and indicate whether the content is likely to be legitimate or suspicious.

4. Objectives

- Develop a basic machine learning approach for phishing and scam detection.
- Identify useful features that can distinguish suspicious content from legitimate content.
- Preprocess and prepare the available dataset for machine learning.
- Train and compare suitable classification algorithms.
- Evaluate the final model using multiple performance measures.
- Create a simple interface for demonstrating the detection process.
- Understand the practical use of machine learning for digital safety.

5. Proposed System

The proposed system accepts message or content-related input, extracts relevant features, and sends the processed information to a trained machine learning classifier. The classifier produces a prediction indicating whether the content is likely legitimate or suspicious. The result is then displayed through a simple application interface. The system is intended as a prototype for educational and demonstration purposes rather than a replacement for professional security systems.

6. Methodology

The project follows a sequence of data preparation, feature extraction, model training, evaluation, testing, and application integration. First, the available examples are reviewed and prepared. Next, useful characteristics are extracted from the messages. The dataset is divided into training and testing portions. Different classification algorithms are trained and compared. After evaluating the results, the most suitable approach is selected for the final prototype. The final system is then tested with different sample inputs.

6.1 Data Preparation

The available dataset contains examples representing legitimate and suspicious content. The data was reviewed for consistency and useful fields were identified. The class labels were prepared so that the machine learning models could learn the difference between the two categories. Difficult or incorrectly classified examples were also reviewed during the improvement stage.

6.2 Feature Extraction

Feature extraction converts message characteristics into information that a machine learning algorithm can use. The project considered suspicious keywords, urgency-related words, number of links, message length, promotional language, unusual URL patterns, excessive punctuation, and abnormal capitalization. Combining several features helps reduce dependence on individual words because a single word may occur in both legitimate and suspicious marketing messages.

7. Machine Learning Algorithms

7.1 Logistic Regression

Logistic Regression is a classification algorithm that estimates the probability of an input belonging to a particular class. It is useful for binary classification problems and provides a relatively simple baseline model for distinguishing legitimate and suspicious messages.

7.2 Decision Tree

A Decision Tree makes predictions through a sequence of feature-based decisions. It is easy to understand because the classification process can be represented as a tree of conditions. It can identify combinations of message characteristics that contribute to a prediction.

7.3 Naive Bayes

Naive Bayes is a probability-based classification method. It is commonly useful for text-related classification because it can work efficiently with features derived from words and message content. It was included to compare a probabilistic approach with the other classification methods.

8. Model Training and Evaluation

The prepared dataset was divided into training and testing data. The training portion was used to build the classification models, while the testing portion was used to check their performance on unseen

examples. The models were compared using accuracy, precision, recall, and F1-score. A confusion matrix was also reviewed to understand correct and incorrect classifications.

Accuracy represents the overall proportion of correct predictions. Precision indicates how often predictions classified as suspicious are actually suspicious. Recall indicates how effectively suspicious examples are identified. The F1-score provides a combined measure of precision and recall. In this project, these measures are more informative than relying on accuracy alone because both missed suspicious messages and incorrectly flagged legitimate messages are important.

9. Final Week Activities

- Reviewed the models and observations obtained during Week 5.
- Finalized the feature set used for the detection process.
- Compared Logistic Regression, Decision Tree, and Naive Bayes results.
- Selected the most suitable approach based on overall evaluation.
- Trained the final model using the prepared features and dataset.
- Performed additional testing with legitimate and suspicious sample messages.
- Reviewed false positives and false negatives to understand model limitations.
- Integrated the prediction process into a simple application/interface.
- Tested the complete workflow from user input to prediction.
- Completed the final project documentation and organized the technical findings.

10. Application / User Interface

A simple application interface was prepared to demonstrate the detection system. The user can provide message or suspicious content as input, and the application passes the information through the required feature-processing steps and trained model. The predicted category is then displayed to the user. The interface is intended to make the machine learning model easier to understand and demonstrate during the final project presentation.

11. Testing

Testing was performed to verify that the application and machine learning workflow operate together correctly. Different legitimate and suspicious examples were used as inputs. The output was checked to ensure that the system generated a classification result. Incorrect predictions were reviewed because they provide useful information about cases where the current feature set or dataset may not be sufficient.

- Input validation was checked before prediction.
- The feature-processing workflow was checked for consistency.
- Legitimate sample messages were tested.
- Suspicious sample messages were tested.
- Prediction output was checked for clarity.
- The complete application workflow was reviewed.

12. Challenges and Solutions

Challenge: Model comparison

Different algorithms can produce different results on the same dataset. I addressed this by comparing multiple evaluation metrics instead of selecting a model based on one result.

Challenge: Similar legitimate and suspicious content

Marketing messages can contain urgency, promotional language, and links even when they are genuine. I therefore considered combinations of features rather than relying only on keywords.

Challenge: Limited dataset

A limited dataset may not represent every type of scam. I reviewed difficult examples and identified the need for more diverse data in future development.

Challenge: Application integration

The feature-processing steps used during training need to be consistent with those used for new inputs. I reviewed the workflow carefully while connecting the model to the application.

13. Final Outcome

The final project resulted in a prototype Phishing and Scam Detection System using machine learning. The project includes data preparation, feature extraction, classification model comparison, evaluation, testing, and a simple application interface. The system demonstrates how multiple message characteristics can be used to support automated identification of potentially suspicious content.

14. Limitations

- The available dataset may not represent all phishing and scam patterns.
- New scam techniques can appear after the model has been trained.
- Some legitimate promotional messages may resemble suspicious content.
- The prototype should not be treated as a guaranteed security solution.
- Detection quality depends on the quality and diversity of training data.

15. Future Enhancements

- Use a larger and more diverse dataset collected from reliable sources.
- Add advanced Natural Language Processing techniques.
- Test additional machine learning and deep learning models.
- Improve analysis of suspicious URLs and shortened links.
- Add multilingual message detection.
- Create a browser extension or mobile application.
- Continuously update the dataset with new phishing and scam patterns.
- Improve the user interface and provide clearer safety guidance.

16. Skills and Knowledge Gained

- Python programming for machine learning and data analysis.
- Dataset preparation and feature engineering.
- Classification model training and comparison.
- Understanding accuracy, precision, recall, F1-score, and confusion matrices.
- Basic application integration with a trained model.
- Problem analysis and interpretation of incorrect predictions.
- Technical documentation and final project presentation.

17. Conclusion

The Phishing and Scam Detection project was completed as a practical machine learning project under the Digital Marketing domain. During the final stage, the work progressed from model improvement to final model selection, testing, application integration, and documentation. The project demonstrated that phishing and scam detection requires more than simple keyword matching because legitimate and suspicious digital communications can share similar characteristics.

Overall, the project provided practical experience in the complete machine learning workflow, from preparing data and extracting features to training, evaluating, testing, and integrating a model into an application. It also improved my understanding of how machine learning can support safer digital communication and digital marketing environments.

18. References

- Python documentation and machine learning learning resources.
- Scikit-learn documentation for classification algorithms and evaluation metrics.
- General educational resources on phishing awareness and online scam detection.
- Project dataset and learning materials used during internship development.